

Raonament Basat en Casos.

Eines de desenvolupament

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CS - FIB- UPC

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Eines CBR Comercials

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CasePower (Induce-it)

(Inductive Solutions)

- **Developer:** Inductive Solutions, NY, USA
- **Platforms:** PC Windows, Mac, OS/2
- **Case organisation:** MS Excel spreadsheet, ordered symbol hierarchies and nested cases
- **Case retrieval:** nearest-neighbour
- **Case adaptation:** via Excel functions and macros
- **Interface:** Excel interface
- **Price approx.:** less than \$ 1000
- **Comments:** Excel must be bought to use this product

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CBR3 (CBR Express) (1)

(Inference Corporation, 1991)

- **Developer :** Inference Corporation, CA, USA
 - <http://www.inference.com>
- **1st Release:** CBR2, 1991. Current CBR3 in 1997
- **Platforms :** PC Windows and MVS version
- **Case organisation:** flat records supporting text and weighted questions
- **Case retrieval:** nearest neighbour and knowledge guided
- **Case adaptation:** no adaptation features
- **Interface:** CasePoint available as a DLL or API and CGI scripts
- **Price approx.:** PC version CBR Express \$ 10000
- **Comments:** CasePoint, Generator, Tester, CasePoint WebServer modules available

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CBR3 (CBR Express) (2)

(Inference Corporation, 1991)

Name	CBRExpress - Inference Corporation
URL	
Data Types	pair <attribute-value>
Attribute Types	Free-form text
Solution (1 o more)	?
Retrieval	Trigrams, heuristic
Indexing	Flat
Adaptation options	?
Documentation	?
Language	Inference's ART-IM expert system toolset and Asymetrix ToolBook
License and price	12000US\$ a long time ago
Source code modification possible?	?

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ESTEEM (1)

(Esteem Software, 1991)

- **Developer** : Esteem Software, IN, USA
- **1st Release**: 1991. Version 1.4, 1997
- **Platforms** : PC Windows. Written in Kappa-PC.
- **Case organisation**: cases can be nested
- **Case Retrieval**: nearest-neighbour with inductive weight generation
- **Case adaptation**: functions and rules
- **Interface**: simple form-based GUI builder
- **Price approx.**: \$ 495
- **Comments**: Version 1.4 supported DDE access and multimedia

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ESTEEM (2)

(Esteem Software, 1991)

Name	ESTEEM - Esteem Software Inc
URL	
Data Types	ASCII file and databases.
Attribute Types	Text, numeric, one of a list, Boolean, case
Solution (1 o more)	Solution: n
Retrieval	Distance: Exact, partial match, word match, range, fuzzy range, equal, inferred (rules) Feature counting, weighted feature counting, inferred feature computation, weighted inferred feature computation
Indexing	ID3 (indices by user)
Adaptation options	Rules by user
Documentation	?
Language	Intellicorp's Kappa-PC and C++
License and price	500\$
Source code modification possible?	?
Comments	Nested Cases / Multi Case base

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KATE tools (Casecraft)

(Acknosoft, 1988)

- **Developer** : AcknoSoft, France
 - <http://www.acknosoft.com>
 - <http://www.kaidara.com> (Kaidara → Servigistics, from 2010 on)
- **1st Release**: 1988.
- **Platforms** : PC Windows and Unix. Written in C
- **Case organisation**: hierarchical cases
- **Case retrieval**: nearest-neighbour and induction
- **Interface**: ToolBook interface can be customised
- **Price approx.**: negotiable
- **Comments**: available as a C library for embedding

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ReCALL (Isoft, 1993)

- *Desenvolupador* : Isoft, France
- *1st Release*: August 1993.
- *Platforms* : PC Windows and Unix. Written in C++
- *Case organisation*: hierarchical cases with relationships
- *Case retrieval*: nearest-neighbour and induction
- *Case adaptation*: demons
- *Interface*: graphical development environment
- *Price approx.*: \$ 9000
- *Comments*: available as a C++ library for embedding

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ReMind (1) (Cognitive Systems, 1992)

- *Desenvolupador* : Cognitive Systems, MA, USA
- *1st Release*: July 1992 v1.1.
- *Platforms* : PC Windows, Mac and Unix. Written in C
- *Case organisation*: nested flat cases and ordered symbol hierarchies
- *Case retrieval*: nearest-neighbour, induction and template
- *Case adaptation*: formulas
- *Interface*: development interface can be customised
- *Price approx.*: \$ 6000
- *Comments*: available as a C library for embedding

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ReMind (2)

(Cognitive Systems, 1992)

Name	ReMind- Cognitive Systems Inc
Data Types	?
Attribute Types	?
Ontologies ...	
Solution (1 o more)	?
Solution can vary	?
Other CBR attributes	?
Retrieval	Nearest neighbour, automatic or guided by user
Indexing	Decision trees and templates
Adaptation options	adaptation formulae (adjustment of difference between new case and retrieved cases)
Documentation	?
Language	C
License and price	6000\$
Is it possible to modify the code	?

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LPA CBR

(Logic Programming Associates, 2004)

Name	LPA CBR
URL	http://www.lpa.co.uk/cbr.htm?gclid=CIX1loeD4KECFR2Y2AodbmXslg
Data	The LPA CBR toolkit assumes a database of previous records, stored in a relational database.
Solution (1 o more)	?
Solution can vary	?
Other CBR attributes	?
Retrieval	No distances: Records are matched by relaxing and/or eliminating parts of the input query until the "threshold" is met. Directly from DataBase
Indexing	No adaptation. Post module of applying rules specified by the user
Documentation	http://www.lpa.co.uk/ftp/4800/cbr_ref.pdf // example and api specification
Language	Prolog
License and price	Copyright © Logic Programming Associates, 2004. All Rights Reserved.
Source code modification possible?	?

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Eines CBR *open-source*

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JColibri

(Grup GAIA, Univ. Polit cnica de Madrid, 2004-)

Name	jColibri
URL	http://gaia.fdi.ucm.es/research/colibri/jcolibri/
CBR Types	Textual, structural, conversational
Attribute Types	Text, attribute-value
Solution (1 o more)	More than one attribute as solution
Solution can vary	?
Other CBR attributes	Result justification and result different from the solution
Retrieval	ExpertClerk Median algorithm. Filters according boolean predicates Lucene (text) Nearest Neighbour Use of weights Flat, index-based (KD tree), ontology-index ?
Indexing	
Missing Treatment	
Adaptation	Copy, direct proportion
Documentation	API documentation
Language	Java
License and price	GNU Lesser General Public License (LGPL)
Source code modification possible?	yes
Extras	Support text, ontologies. Connection to database using hibernate. Includes metrics for evaluating the case base (including visualization), extension for the construction of case-based recommender systems, evolution for the explanation of intelligent Systems with CBR

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eXIT*CBR

(Control Engineering and Intelligent Systems (eXIT), Univ. Girona,, 2008-)

Name	eXITCBR
URL	http://exitcbr.udg.edu/
CBR Types	?
Attribute Types	Attribute-value
Solution (1 o more)	?
Solution can vary	?
Other CBR attributes	Structural CBR developed for medical medical systems
Retrieval	measure for unknown attributes, string measure (Hamming distance), numeric measures (Hamming and Euclidean distance), boolean measure (equals), date measure (distance-based), sequence-measures (e.g., distance-based), aggregation measures (e.g., Euclidean, mean)
Indexing	Flat
Missing Treatment	?
Adaptation	Copy, probabilistic methods for binary cases, majority methods, probabilistic multi-class methods
Documentation	API documentation, Tutorial
Language	Java
License and price	?
Source code modification possible?	Source code not publicly available, sharable upon request
Extras	Provides plugin capabilities for special purpose: genetic algorithms (for feature learning), family risk calculator to manage GEDCOM data. Since version 2.0 supports Distributed CBR as a Multiagent System.

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ProCAKE

(Dept. of Business Information Systems II, Univ. of Trier, Germany, 2005-)

Name	ProCAKE
URL	https://procake.uni-trier.de/
CBR Types	Textual, structural, process-oriented
Attribute Types	Text, attribute-value, object, (semantic) graphs, collections(list, set), custom data classes (XML, Java)
Solution (1 o more)	?
Solution can vary	?
Other CBR attributes	Possibility to write own similarity measures (Java)
Retrieval	generic measures (table-based, equals), string measures (e.g., Levenshtein, taxonomy-based), numeric and date measures (e.g., distance, Sigmoid), sequence measures (e.g., DTW, SWA), interval measure (distance-based), ontology measures (e.g., path-based, equivalence), graph/process measures (e.g., DTW, SWA, Levenshtein, A*-based mapping), aggregation measures (e.g., average, maximum, minimum, Minkowski)
Indexing	Flat
Missing Treatment	?
Adaptation	Null adaptation, integration of custom adaptation methods supported
Documentation	Wiki site
Language	Java
License and price	?
Source code modification possible?	?
Extras	Domain-independent adaptation manager

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CloudCBR

(AI and Reasoning Group, Robert Gordon University Aberdeen, 2020-)

Name	CloudCBR
URL	https://cloudcbr.com/
CBR Types	Textual, structural
Attribute Types	Text, attribute-value, object
Solution (1 o more)	?
Solution can vary	?
Other CBR attributes	Supports building distributed, highly scalable, and generic CBR systems based on a microservices architecture
Retrieval	generic measure (equals), string measures (e.g., BM25 measure, word embedding vector-based), numeric measures (e.g., interval, nearest number), categorical measures (distance, table), date measure (nearest date), location measure (nearest location), ontology measures (path-based, feature-based), aggregation measure (weighted sum)
Indexing	Flat
Missing Treatment	?
Adaptation	Null adaptation, supports custom adaptation via scripts
Documentation	API documentation
Language	Java, Python, JavaScript
License and price	?
Source code modification possible?	?
Extras	Supports integration of custom similarity measures via scripts. Supports manual adjustment of values via GUI.

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freeCBR

(Lars Johnson, 2003-2018)

Name	freeCBR
URL	http://freecbr.sourceforge.net
Data Type:	List <attribute, value>
Attribute Types	Integer, boolean, float, string, multi string
Solution (1 o more)	?
Solution can vary	?
Retrieval	Euclidean
Weights	Supported
Missing Values	Ignored
Indexing	Flat
Adaptation	Not implemented
Documentation	Only API documentation
Language	Java
License and price	Open, even to sell
Source code	Yes
modification possible?	

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myCBR

(German Research Center for Artificial Intelligence - DFKI , 2007-)

Name	myCBR
Url	http://mycbr-project.org/
CBR Type	Textual, Structural
Attribute Type	Numerical, Symbolic, Text, Set
Solution (1 o more)	1
Other CBR attributes	?
Retrieval	generic measures(table-based,equals), string measures (e.g., Levenshtein, taxonomy-based), numeric and date measures (e.g., distance, Sigmoid), sequence measures(e.g.,DTW, SWA), interval measure (distance-based), ontology measures (e.g.,path-based, equivalence), graph/process measures(e.g.,DTW, SWA, Levenshtein, A*-basedmapping), aggregation measures(e.g.,average, maximum, minimum, Minkowski)
Indexing	Flat
Adaptation options	Null adaptation, integration of custom adaptation methods supported
Documentation	Tutorial
Language	Java
License and price	open source, developed under the GPL license.
Source code modification possible?	Yes
Comments	Standalone app or Protégé plugin Scriptable similarities in Jython

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CASPIAN

(Centre for Intelligent Systems, Univ. of Wales, 199?)

Name	Caspian
URL	http://www.aber.ac.uk/~dcswww/Research/mbmsg/cbrprojects/getting_caspian.shtml
Data Types	Number, string
Attribute Types	Symbol, number, string, list
Solution (1 o more)	Solution: n
Solution can vary	?
Other CBR attributes	Evaluation
Indexing	Flat
Retrieval	It is possible to skip some attributes but not those that are used for indexing. Uses weights, special similarity function
Indexing	Attribute indices to start the search
Adaptation options	Modification Rules (= ESTEEM) by user
Documentation	Casl and Caspian manual
Language	Casl for defining cases and the database
License and price	the program is licensed free of charge, there is no warranty for the program, to the extent permitted by applicable law
Source code modification possible?	yes

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IUCBRF

(Indiana University, 2005)

Name	IUCBR
Url	https://homes.luddy.indiana.edu/leake/iucbrf/index.html
Data Types	
Attribute Types	Double, integer, boolean, set, term vector, internet address. Support extending new attributes types
Solution (1 o more)	Support more than 1 attribute as solution
Solution can vary	?
Other CBR attributes	Inactive context, use counts, time of creation, source
Retrieval	It is possible to skip some attributes but not those that are used for indexing. Uses weights, special similarity function
Indexing	Flat or B-tree-backed (selected features by user)
Adaptation options	Copy, weighted average and weighted majority
Documentation	A tutorial
Language	Java
License and price	Open Software License
Source code modification possible?	Yes
Extra	Conversational CBR

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AIAI CBR Shell

(Artificial Intelligence Applied Institute, Univ. of Edinburgh, 1990-2019)

Name	AIAICBR
Url	http://www.aiai.ed.ac.uk/project/cbr/CBRDistrib/
Data Types	Numerical, string
Attribute Types	Numerical, string
Solution (1 o more)	1 nominal or discrete solution
Solution can vary	?
Other CBR attributes	?
Retrieval	Nearest Neighbour
	Numerical comparison and strings by trigrams
	Weights can be optimized by genetic algorithms
Indexing	Flat
Adaptation options	?
Documentation	Scarce
Language	Java and C++
License and price	?
Source code modification possible?	?

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Case Base Generator (CaBaGe)

(University of Novi Sad, 2008)

Name	CaBaGe
URL	
Data Types	set of values of some selected,
Attribute Types	most important attributes.
Solution (1 o more)	Integer, float, string
Solution can vary	?
Other CBR attributes	?
Retrieval	Case Retrieval Net (CRN)
	Support weights
Indexing	CRN
Adaptation options	None
Documentation	Two papers, no more information neither the code
Language	Java
License and price	?
Source code modification possible?	?

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Gesconda-CBR

(KEMLG, IDEAI-UPC, UPC, 2014)

Name	GESCONDA- CBR
URL	
Data Types	Instance Based
Attribute Types	Continuous and Discrete {Ordered and non-ordered}
Solution (1 o more)	Number of solutions can be more than one
Solution can vary	Solution attributes can vary
Other CBR attributes	Evaluation, Utility
Retrieval	Nearest Neighbour
	Customizable distance:
	Implemented ones: Minkowsky, L'Eixample, Canberra, Clark, Cosinus.
	Support weights (possibility automatic algorithms)
Indexing	Flat memory, Discrimination tree, SOM
Adaptation options	Copy, mean, weighted Mean (weight = distance, other parameter), user formulas
Use of evaluation or utility	It is possible
Documentation	Up to now, the Master Thesis as a manual and as technical report and javadoc
Language	Java
License and price	Free for academic purposes when stable
Source code modification possible?	Yes and easily extendible
Comments	An adaptation without interface has been done for the ModSimTex project.

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